09-Hunt-to-Detection-Pipeline-Checklist

Hunt-to-Detection Pipeline Checklist

Use this checklist to convert a threat hunt finding into a production detection rule. This pipeline applies to findings from purple team exercises, ad-hoc hunts, CTI-driven hunts, or anomaly investigations. Complete each phase before moving to the next.

Finding Information

Field	Value
Hunt ID / Ticket	
Hunt Analyst	
Date Finding Identified	
Trigger Source	Purple Team Exercise / CTI Report / Anomaly / Ad-hoc Hunt
ATT&CK Technique ID	
Technique Name	
Summary of Finding	

Phase 1: Hypothesis Formation

- ❑ Define the hunt hypothesis in a single testable statement
  - Template: "An adversary performing [technique] will generate [observable] in [data source] that can be distinguished from legitimate activity by [differentiator]."
  - Example: "An adversary using Pass-the-Hash will generate Event ID 4624 with Logon Type 9 from a non-standard source process."
- ❑ Identify the trigger that initiated this hunt:
  - ☐ Purple team exercise gap (reference exercise ID and TTP)
  - ☐ New CTI report (reference report or intel source)
  - ☐ Detection gap from previous exercise

- [ ] Anomalous behavior observed during monitoring
- [ ] Other: _____
- [ ] Map the hypothesis to an ATT&CK technique and tactic
- [ ] Identify the data sources needed to test the hypothesis (reference ATT&CK Data Sources)
- [ ] Confirm the required data sources are available and ingesting

Methodology reference: PEAK Framework (Prepare, Execute, Act, Knowledge) or SANS Hypothesis-Driven Hunting

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Phase 2: Query Development

- [ ] Write an initial query against available telemetry to test the hypothesis
- [ ] Document the query details:
  - [ ] Platform (SIEM, EDR console, hunting tool)
  - [ ] Query language (SPL, KQL, EQL, Lucene, SQL)
  - [ ] Data sources and index/table names
  - [ ] Key fields and filtering logic
  - [ ] Time range searched
- [ ] Run the query against historical data (minimum 30 days where possible)
- [ ] Record the results:
  - [ ] Total events returned
  - [ ] True positives identified (if any)
  - [ ] False positives / benign matches
  - [ ] No results (refine hypothesis or query)
- [ ] Iterate on the query if initial results are too noisy or too narrow

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Phase 3: Validation and Refinement

- [ ] If the hunt was triggered by a purple team exercise or CTI report, coordinate with the Red Team to execute the TTP in a controlled setting
- [ ] Confirm the query detects the TTP execution
- [ ] Execute variations of the TTP to test query breadth:
  - [ ] Different tools or utilities
  - [ ] Different parameters or flags
  - [ ] Different user contexts (standard user, admin, service account)
  - [ ] Different target systems (workstation, server, cloud)
- [ ] Refine the query to distinguish malicious behavior from legitimate activity
- [ ] Document the differentiators (what makes this malicious vs. benign)
- [ ] Measure the false positive rate against baseline activity
- [ ] Record sample events: at least 2 true positive examples and 2 false positive examples (if any)

Gate: The query must reliably detect the TTP with a manageable false positive rate before proceeding. If the query cannot distinguish malicious from benign, document why and close the finding as "not automatable" (it remains a hunt playbook only).

Phase 4: Detection Candidate Submission

❑ Package the finding as a detection candidate with the following:

Field	Value
ATT&CK Technique ID	
Technique Name	
Tactic	
Query Logic	(paste or link to query)
Data Source Requirements	
Target Platform	SIEM / EDR / NDR / Cloud-native
Expected False Positive Rate	
Sample True Positive Events	(attach or link)
Sample False Positive Events	(attach or link)
Recommended Alert Severity	Critical / High / Medium / Low
Robustness Assessment	(Pyramid of Pain level or Summiting the Pyramid score)
Hunt Analyst Notes	

❑ Submit the candidate to the detection engineering backlog (JIRA, GitLab Issues, shared backlog, or equivalent)
❑ Tag the submission with the source trigger (exercise ID, CTI report, hunt ID)

Phase 5: Detection Engineering Review and Deployment

This phase is owned by the Detection Engineering team. The hunt analyst supports as needed.

❑ Detection engineer reviews the candidate and accepts or returns with questions

- [ ] Author the detection rule in the appropriate format:
  - [ ] Sigma 2.0 (preferred for portability)
  - [ ] Platform-native rule
  - [ ] YARA (for file/memory patterns)
- [ ] Follow the Detection Lifecycle Checklist from Phase 2 (Design) onward:
  - [ ] Peer review
  - [ ] Validation against TTP execution
  - [ ] False positive tuning
  - [ ] Production deployment
  - [ ] Regression test enrollment
- [ ] Create or update the SOC response playbook for this detection

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Phase 6: Knowledge Base Update

- [ ] Add the hunt finding and resulting detection to the team's knowledge base or wiki
- [ ] Update the ATT&CK Navigator coverage layer to reflect the new detection
- [ ] If the finding originated from a purple team exercise, update the exercise tracking system (PlexTrac, VECTR, JIRA) to mark the gap as addressed
- [ ] If the finding originated from a CTI report, link the detection back to the intelligence source
- [ ] Share the finding at the next joint Hunt / Detection Engineering review session
- [ ] Close the original hunt ticket with a summary:
  - [ ] Hypothesis (confirmed / refuted / partially confirmed)
  - [ ] Detection rule name and ID
  - [ ] Production deployment date
  - [ ] Regression test schedule

Handoff Tracking

Milestone	Date	Owner
Hypothesis formed		Hunt Analyst
Query validated		Hunt Analyst
Candidate submitted to backlog		Hunt Analyst
Detection engineer assigned		Det. Eng. Lead
Rule deployed to production		Detection Engineer
Regression test confirmed		Detection Engineer
Knowledge base updated		Hunt Analyst
Hunt ticket closed		Hunt Analyst

Total Pipeline Duration: ___ days (target: < 30 days from validated finding to production detection)
